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mAIstro: An open-source multi-agent system for automated end-to-end development of radiomics and deep learning models for medical imaging

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ABSTRACT

Objective: To develop an autonomous, open-source, multi-agent system capable of performing end-to-end artificial intelligence (AI) workflows in medical imaging, including exploratory data analysis, feature importance analysis, radiomic feature extraction, and the development and deployment of segmentation, classification, and regression models.

Materials and methods: The system comprises a master agent that coordinates multiple task-specific agents, each responsible for executing a distinct AI function. Based on natural language prompts, the system performs reasoning, selects the appropriate tools, and carries out complex, multi-step workflows autonomously. The system's performance was evaluated using 20 open-source datasets spanning structured clinical data and multimodal medical imaging across a range of anatomical regions and pathologies. Evaluation included a diverse set of task and workflow prompts. Output correctness was confirmed by monitoring system logs for agent and tool behavior and by re-executing all tools manually to ensure identical results.

Results: The system successfully executed all tasks when powered by high-end large language models (LLMs), demonstrating robust performance across exploratory and feature importance analysis, radiomic extraction, segmentation, and predictive modeling. At no point did the user instruct mAIstro which task-specific agents to invoke or how to solve the task. mAIstro autonomously understood the prompt, devised a strategy, selected the appropriate agents, and coordinated their execution to complete complex workflows without human intervention. All outputs matched those produced through manual tool execution.

Conclusion: This study introduces an LLM-agnostic, multi-agent framework for reproducible and autonomous development of medical AI pipelines. Code and implementation instructions are available at: https://github.com/eltzanis/mAIstro.

Introduction

The rapid evolution of large language models (LLMs) has influenced multiple areas in healthcare, including clinical support, patient communication, and medical education [1,2]. LLMs have shown capabilities in translating clinical notes, summarizing medical reports, and supporting diagnostic reasoning [3–5]. In radiology and nuclear medicine, they have been tested for report generation, protocol optimization, and structured interpretation assistance [3–6]. Early evaluations suggest that LLMs may enhance efficiency and consistency in clinical workflows [6,7].

However, LLMs cannot autonomously interact with their environment, retrieve or process new datasets, execute code, or interface with external pipelines, limiting their integration in real-world biomedical

Abbreviations: LLM, Large Language Model; AI, Artificial Intelligence; EDA, Exploratory Data Analysis; AUC, Area Under the Curve; MAE, Mean Absolute Error; MSE, Mean Squared Error; RMSE, Root Mean Squared Error; RMSLE, Root Mean Squared Log Error; MAPE, Mean Absolute Percentage Error; SD, Standard Deviation; DSC, Dice Similarity Coefficient; IoU, Intersection over Union.

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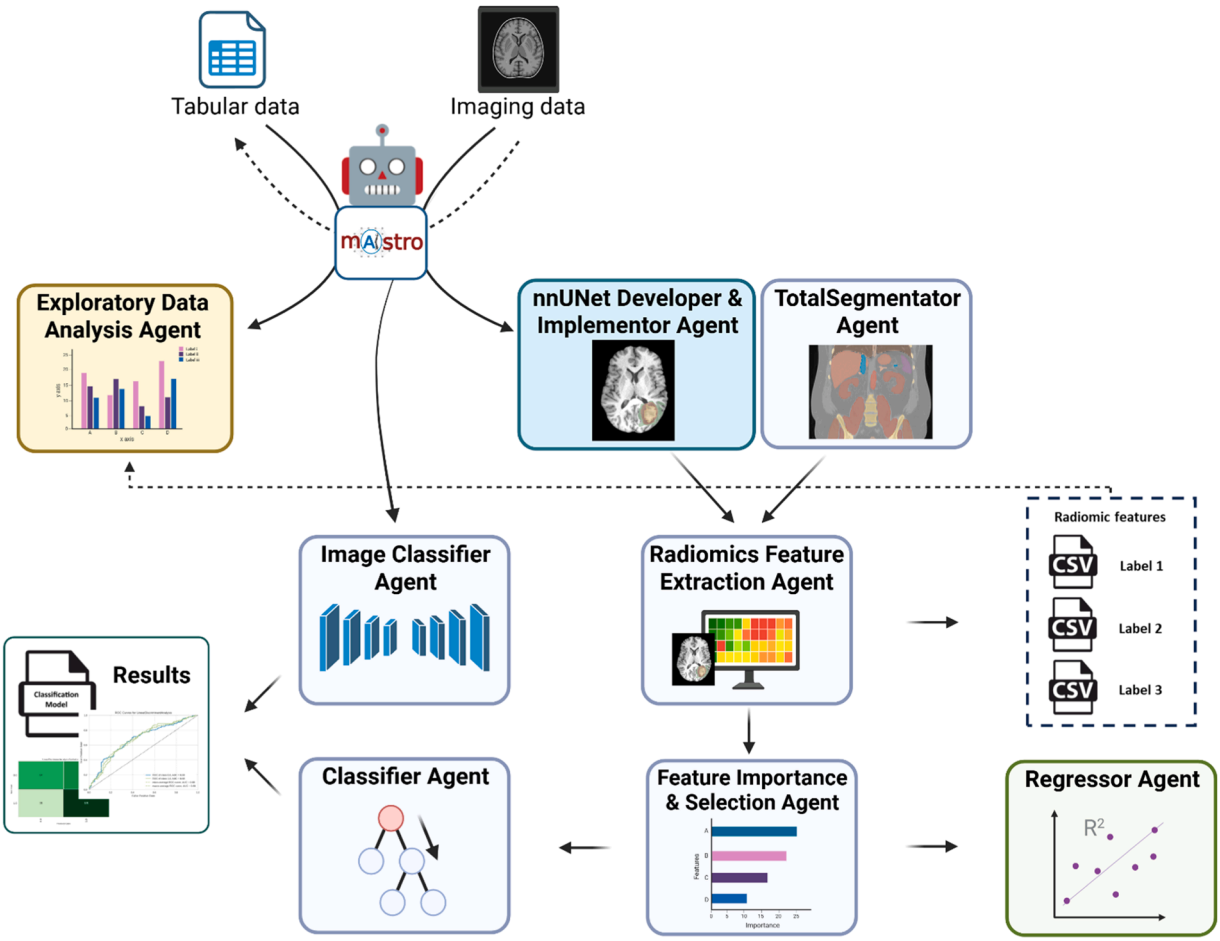

Fig. 1. Architecture of the developed agentic system for autonomous medical data analysis and AI modeling. The mAIstro master agent orchestrates a team of eight specialized task-specific agents, each designed to autonomously execute distinct functions. Task-specific agents include modules for exploratory data analysis, feature importance analysis, radiomics feature extraction, segmentation model development, classification and regression model development based on tabulated data, and image classification model training. Indicative interactions between agents are presented with arrows (created with biorender.com).

settings [8,9]. Agentic systems, built on cognitive reasoning architectures, offer a solution. These frameworks enable LLMs to reason, plan, and execute tasks through iterative cycles of thought, action, and observation using external tools. Prompting strategies such as ReAct [10], Chain-of-Thought [11], and Tree-of-Thoughts [12] form the foundation of these systems, allowing LLMs to solve complex problems through structured reasoning [13].

Despite progress in AI, challenges remain in reproducibility, transparency, fairness, and generalizability. Many published AI models are not transferable to clinical or research settings due to non-standardized development processes or limited accessibility. Additionally, users without programming skills are often excluded from training, evaluating, or deploying AI tools [14].

To address these challenges, our objective was to develop an open-source, autonomous, multi-agent system capable of understanding and executing natural language instructions for complex biomedical data tasks. Each agent within the framework is designed for a specific task, and the agents interact in coordination to complete end-to-end workflows. The proposed agentic framework is designed to promote accessibility, standardization, and methodological rigor across clinical and research AI workflows. It supports a wide range of tasks such as exploratory data analysis, radiomic feature extraction, image and tabular data modeling, and deployment of state-of-the-art segmentation and classification pipelines, without requiring user interaction beyond natural language prompts.

Materials and methods

Multi-agent system

The agentic system was developed using the smolagents library [15]. The system consists of a master agent, which orchestrates a team of eight task-specific agents. Each agent is designed to perform a specialized task (Fig. 1). These tasks include exploratory data analysis (EDA), radiomic feature extraction, feature importance analysis, and the development and implementation of segmentation, classification and regression models. The functionality of each agent, along with the tools and methods employed, is described in the Supplementary Material (Part A). The system was developed on a workstation equipped with an AMD Ryzen 7 5800 × 8-core CPU, 64 GB RAM, and an NVIDIA RTX 2080 Ti GPU with 11 GB memory.

Datasets

Ethics committee approval was not required, as all data used in this study are publicly available and designated for open research use. A diverse collection of datasets was used in this study, including Breast Cancer Wisconsin, Heart Disease, Heart Failure, Pima Indians Diabetes, Life Expectancy (WHO), BraTS 2021, KiTS23, MAMA-MIA, and several image classification datasets from the MedMNIST collection [16–25]. A detailed description of each dataset is provided in Supplementary Material (Part B).

Table 1
Query success rates across LLMs and tasks.

LLM	Image Classification (Training)	Image Classification (Inference)	Regression (Training)	Regression (Inference)	Classification (Training)	Classification (Inference)	TotalSegmentator	nnUNet	Feature Importance	EDA	Radiomics	Success Rate (%)
GPT−4o	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
GPT−4.1	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
Sonnet 3.7	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
DeepSeek V3	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
DeepSeek R1	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
Llama 3.3 70B	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	100
Llama 4 17B	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Pass	Pass	Pass	91
Llama 3.1 8B	Fail	Fail	Fail	Fail	Pass	Fail	Fail	Fail	Fail	Pass	Fail	18
Mistral 24B	Pass	Fail	Fail	Fail	Pass	Fail	Fail	Fail	Pass	Pass	Fail	36
DeepSeek R1 14B	Pass	Pass	Fail	Fail	Fail	Fail	Pass	Pass	Fail	Pass	Pass	55
Mistral 7B	Fail	Fail	Fail	Fail	Fail	Fail	Pass	Fail	Fail	Fail	Fail	10
QwQ 32B	Pass	Pass	Fail	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	90

Experimental setup

To evaluate the functionality and robustness of the developed agentic framework, a comprehensive series of queries were executed. These queries were constructed to test two aspects:

i. whether the master agent correctly identifies and deploys the appropriate task-specific agent based on the task requirements described in the query, and
ii. whether the deployed agent successfully performs the assigned task in a fully autonomous and correct manner.

A large and diverse collection of queries was created, covering a wide spectrum of tasks, including exploratory data analysis, feature importance analysis, classification model development and inference, regression model development and inference, radiomic feature extraction, medical image segmentation, and image classification model development and inference. All tested queries are provided in the Supplementary Material (Part C).

In addition to single-task queries, complex multi-task queries were also constructed to challenge the system's ability to sequentially deploy multiple agents within a single workflow. These included tasks such as segmenting organs, extracting radiomic features from the segmented regions, performing EDA on the extracted features, and developing classification models from the processed datasets.

All experiments were executed in a fully automated manner, with no manual intervention after query submission, thereby validating both the agent selection mechanisms and the task-specific tool execution pipelines of the proposed framework.

To ensure the correctness of the agentic outputs, we employed a two-pronged verification process. First, we monitored the system logs during execution to confirm that the correct agents and tools were activated in response to each query, and that the inputs passed to the tools aligned precisely with the instructions in the prompt. Second, for each task, the corresponding tool was executed independently by the developer using the same input parameters. The outputs generated by the manually executed tools were then compared to those produced by the agentic system.

To assess the impact of the utilized LLM on the efficiency and reliability of the agentic system, all queries were re-executed using different LLMs as the core reasoning engine. Specifically, the following LLMs were tested: GPT-4o, GPT-4.1, Claude Sonnet 3.7, DeepSeek V3, DeepSeek R1, Llama 3.3 70B, QwQ 32B, Mistral 24B, DeepSeek R1 14B, Llama 4 Scout 17B, Llama 3.1 8B and Mistral 7B. The consistency of agent selection and task execution success rates were recorded for each LLM configuration.

Results

System performance across large language models

Table 1 summarizes the success rates across task types and LLMs. A task was considered successful if the master agent correctly identified the required agent, activated it, and the agent autonomously completed the task without intervention or critical errors. Tasks included exploratory data analysis, feature importance analysis, classification model development and inference (tabulated data), regression model development and inference (tabulated data), radiomic feature extraction, medical image segmentation, and image classification model development and inference.

High-performing LLMs, including GPT-4o, GPT-4.1, Claude Sonnet 3.7, DeepSeek V3, DeepSeek R1, and Llama 3.3 70B, achieved a 100 % task success rate across all tested tasks. Moderate performance was observed with Llama 4 Scout 17B (91 % success rate) and QwQ 32B (90 % success rate). In contrast, smaller models such as Llama 3.1 8B, Mistral 24B, Mistral 7B, and DeepSeek R1 14B demonstrated substantially lower success rates, ranging from 10 % to 55 %. These results

Table 2
Classification model results.

Dataset	Accuracy (%) (SD)	AUC (SD)	Recall (SD)	Precision (SD)	F1-score (SD)
Breast Cancer	97.2 (2.1)	0.998 (0.005)	97.2 (2.1)	97.4 (2.0)	97.2 (2.1)
Heart Disease	85.4 (5.7)	0.925 (0.049)	90.5 (7.1)	84.6 (7.8)	87.1 (5.0)
Heart Failure	73.2 (7.7)	0.762 (0.085)	67.6 (24.0)	59.1 (10.4)	60.3 (12.8)
Diabetes	76.2 (6.0)	0.831 (0.063)	68.5 (7.7)	66.2 (9.5)	66.9 (6.7)

SD = Standard Deviation, AUC = Area Under the Curve

Table 3
Regression model results (Life Expectancy Dataset).

Metric	Mean (SD)
MAE	0.242 (0.016)
MSE	0.131 (0.025)
RMSE	0.361 (0.035)
R^2	0.9985 (0.0004)
RMSLE	0.0060 (0.0008)
MAPE	0.0038 (0.0003)

MAE = Mean Absolute Error, MSE = Mean Squared Error, RMSE = Root Mean Squared Error, RMSLE = Root Mean Squared Log Error, MAPE = Mean Absolute Percentage Error, SD = Standard Deviation

highlight the trade-offs between model scale and practical deployment: larger models offer high accuracy and reliability but demand greater computational resources, while smaller models are faster and less costly to run but fail to function as reliable reasoning engines for complex agentic frameworks. Model choice therefore depends on balancing resource availability with the robustness required for biomedical applications.

Evaluation on tabulated data

EDA and feature importance agents

All queries related to EDA and feature importance analysis were successfully completed across all tested tabulated datasets. The EDA Agent generated comprehensive summary reports, multiple types of visualizations (e.g., histograms, correlation matrices, pair plots), and textual insights regarding dataset characteristics. All plots were saved correctly to the specified output directories as requested. The Feature Importance Analysis Agent successfully performed feature ranking. Top-ranked feature sets were saved in structured CSV files as requested, and diagnostic plots were generated accordingly. No errors or incomplete outputs were observed during any EDA or feature selection tasks. Representative results and visual outputs from these analyses are provided in Supplementary Material Part D.

Classifier agent

Blended classification models were developed for four tabulated datasets: Breast Cancer Wisconsin, Heart Disease, Heart Failure, and Pima Indians Diabetes datasets [16–19]. Each blended model was constructed through the following steps: training of multiple baseline models, selection of the top three models based on cross-validation performance, hyperparameter tuning of the selected models, creation of a final blended ensemble combining the three tuned models. The resulting evaluation metrics, including accuracy, Area Under the Curve

Table 4
Image classification model performance.

Dataset (Image size)	Model	Accuracy (%)	Macro-averaged Precision (%)	Macro-averaged Recall (%)	Macro-averaged F1-score (%)
PneumoniaMNIST (28 ×28)	ResNet18	89.1	91.9	85.7	87.6
	ResNet50	87.8	91.8	83.8	85.9
	InceptionV3	88.9	92.0	85.4	87.4
	VGG16	86.1	90.4	81.6	83.7
PathMNIST (28 ×28)	ResNet18	89.1	90.2	87.8	87.1
	ResNet50	90.4	89.8	89.2	88.7
	InceptionV3	88.5	89.1	88.3	87.4
	VGG16	91.1	90.5	89.4	87.4
PathMNIST (64 ×64)	ResNet34	89.7	90.9	87.3	87.7
BreastMNIST (28 ×28)	ResNet18	85.3	82.0	79.4	80.5
	ResNet50	86.5	83.3	81.8	82.5
	InceptionV3	85.9	86.9	76.1	79.3
	VGG16	87.2	86.1	79.9	82.3
BreastMNIST (128 ×128)	ResNet50	88.5	86.7	83.1	84.6
DermaMNIST (28 ×28)	ResNet18	78.6	69.0	55.9	60.3
	ResNet50	77.7	62.1	51.9	55.3
	InceptionV3	77.7	62.2	52.3	55.4
	VGG16	75.8	70.2	44.3	47.7
DermaMNIST (224 ×224)	ResNet101	79.6	59.2	61.1	59.8
OrganAMNIST (28 ×28)	ResNet18	94.9	94.8	94.4	94.5
	ResNet50	95.1	95.3	94.6	94.9
	ResNet152	94.4	94.7	93.9	94.2
	VGG16	95.1	94.9	94.5	94.7
OCTMNIST (28 ×28)	ResNet18	77.8	82.3	77.8	75.5
	ResNet50	79.2	83.2	79.2	77.2
	InceptionV3	79.4	83.6	79.4	77.3
	VGG16	79.5	83.7	79.5	77.2
BloodMNIST (28 ×28)	ResNet18	97.8	97.7	97.6	97.6
	ResNet50	97.4	97.3	97.2	97.2
	InceptionV3	97.8	97.8	97.9	97.8
	VGG16	97.4	97.3	97.0	97.1
BloodMNIST (128 ×128)	InceptionV3	98.9	99.1	99.0	99.1

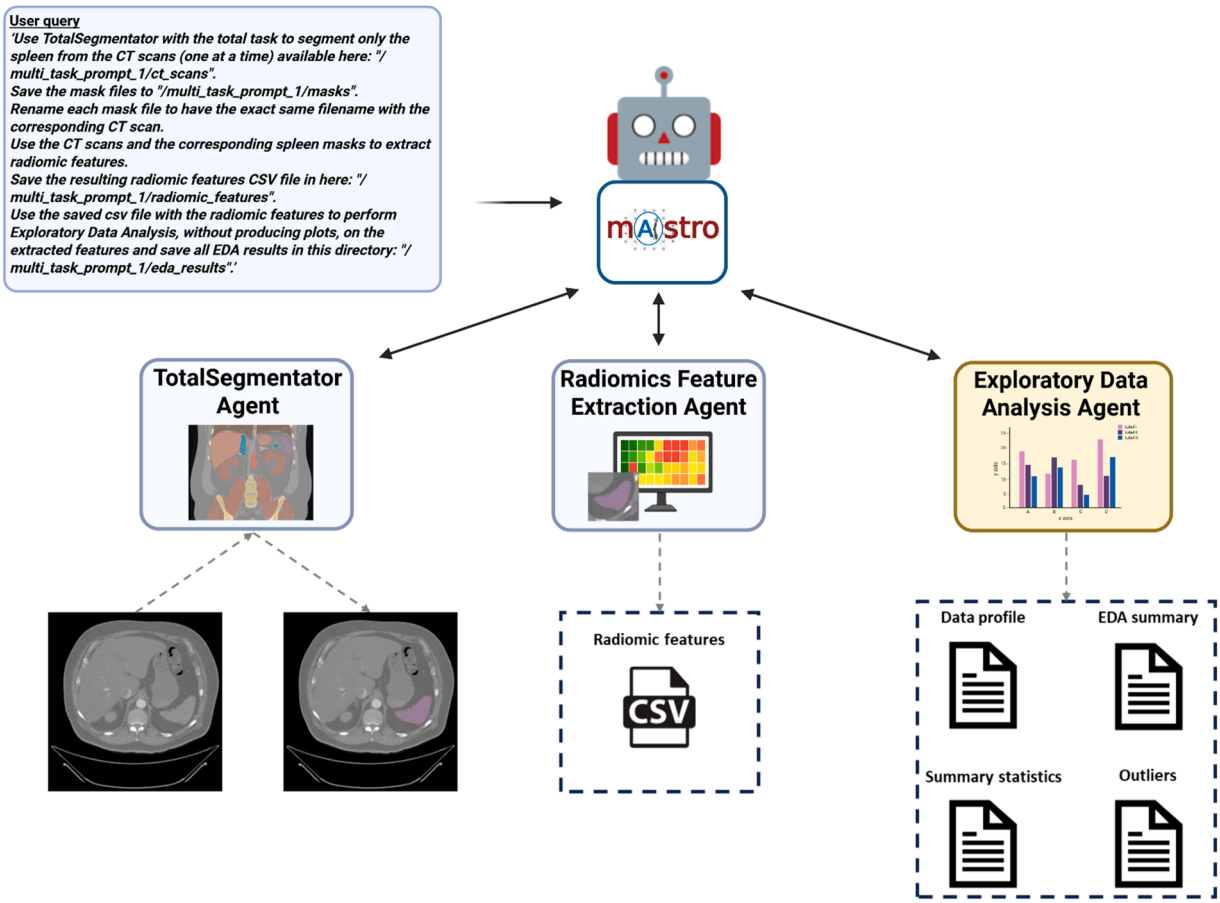

Fig. 2. Workflow execution for multi-task prompt 1, involving a sequence of agent invocations coordinated by the mAIstro. The TotalSegmentator Agent segments the spleen from input CT scans; the resulting masks are used by the Radiomic Feature Extraction Agent to extract quantitative features. These features are then processed by the EDA Agent (created with biorender.com).

(AUC), recall, precision and F1-score are presented in Table 2. AUC provides a measure of discriminative ability across all possible probability thresholds, whereas accuracy reflects performance at a single chosen probability threshold (commonly 0.5) and is therefore sensitive to both threshold selection and class prevalence. Precision, recall, and F1-score capture the balance between false positives and false negatives, offering greater insight under conditions of class imbalance [26]. Because the most suitable metric depends on the clinical scenario, we designed the agentic system to report multiple metrics by default. This ensures that users have all key measures available and can select the most appropriate one to interpret the model's performance in their specific context.

Regressor agent

For the regression task on the Life Expectancy dataset [20], a blended model was similarly developed following model comparison and tuning procedures. The evaluation metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), Root Mean Squared Log Error (RMSLE), and Mean Absolute Percentage Error (MAPE), are summarized in Table 3.

Evaluation on medical image data

nnU-Net developer and implementer agent

The nnU-Net Developer and Implementer Agent was evaluated using the BraTS 2021 (multimodal brain MRI) [23] and KiTS23 (abdominal CT) datasets [24]. For each dataset, a 3D full resolution UNet model was trained and validated using the autonomous agentic workflow.

Evaluation was based on standard segmentation performance metrics, including Dice similarity coefficient (DSC) and Intersection over Union (IoU), calculated on the validation sets. For the BraTS 2021 dataset [23], whole tumor segmentation achieved a mean DSC of 0.96 and an IoU of 0.92. Tumor core segmentation achieved a mean DSC of 0.95 and an IoU of 0.91. Enhancing tumor segmentation achieved a mean DSC of 0.89 and an IoU of 0.83.

For the KiTS23 dataset [24], kidney segmentation achieved a mean DSC of 0.95 and an IoU of 0.91 while kidney tumor segmentation achieved a mean DSC of 0.74 and an IoU of 0.64.

These results demonstrate that the nnU-Net Agent successfully managed the full training and evaluation workflow for 3D medical image segmentation, achieving high segmentation accuracy on clinically relevant anatomical structures and tumors.

TotalSegmentator agent

The TotalSegmentator Agent was evaluated using images from the BraTS 2021 [23] and KiTS23 [24] datasets. Tasks included full-body segmentation and organ-specific segmentation requests on CT and MRI images. The agent successfully performed segmentation in all queries, generating and saving binary masks for the requested organs and tissues as NIfTI files.

Segmentation outputs were visually inspected and verified for correctness in anatomical location and correspondence to the requested structures. No task failures or incomplete segmentations were observed across the full evaluation set.

Radiomic feature extraction agent

The Radiomic Feature Extraction Agent was evaluated on three

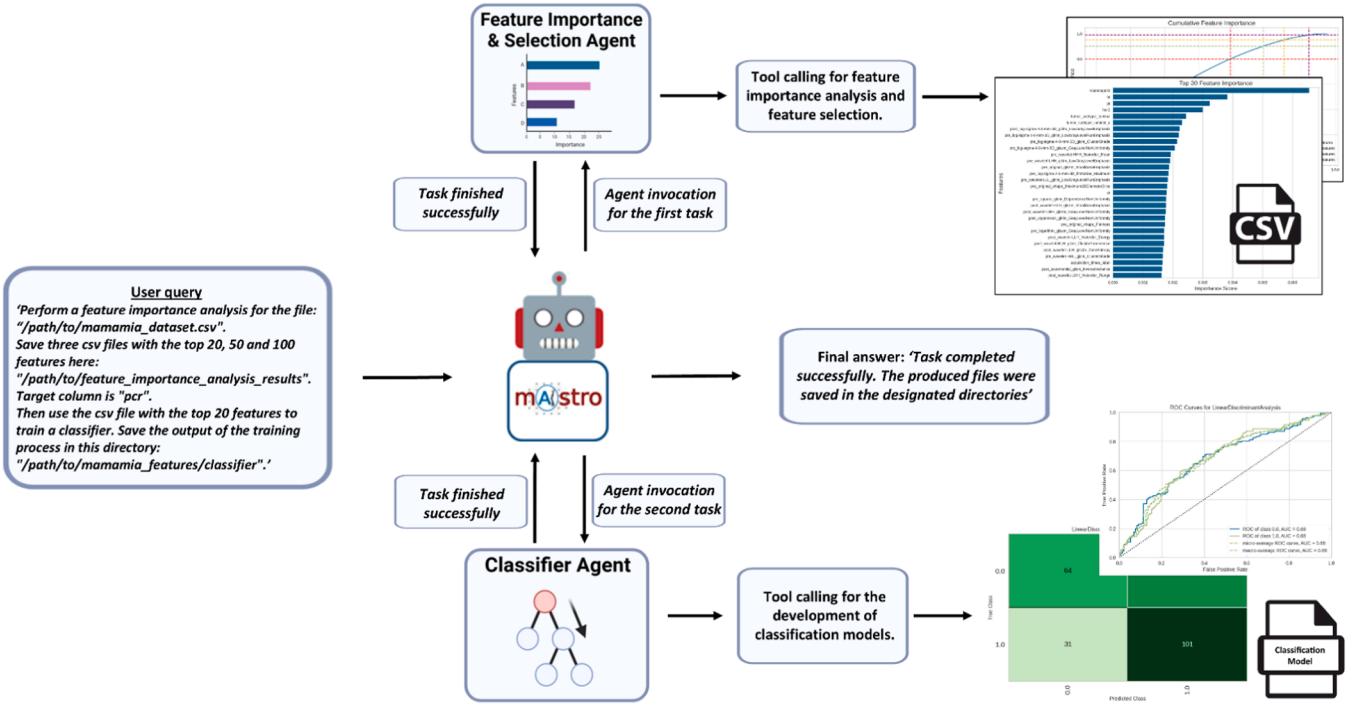

Fig. 3. Multi-agent sequential workflow for real-world clinical data processing based on the MAMA-MIA dataset. A user query requesting sequential feature importance analysis and classifier training is parsed by the master agent, which autonomously invokes the appropriate specialized agents. First, the Feature Importance and Selection Agent identifies top-ranked predictive features from the provided CSV file, focusing on the target variable "pcr" (pathological complete response). The resulting features are saved and used to initiate a second task, where the Classifier Agent trains a predictive model. Throughout the workflow, each agent autonomously selects and executes the appropriate tool, saves intermediate outputs and produces final classification results (created with biorender.com).

medical imaging datasets: BraTS 2021 (brain multiparametric MRI) [23], KiTS23 (abdominal CT) [24], and MAMA-MIA (breast DCE-MRI) [25]. Radiomic features were extracted across multiple test scenarios, including, extraction from individual segmentation labels (e.g., tumor core, whole kidney), use of different image filters (e.g., original images, wavelet-filtered, Laplacian of Gaussian), selection of specific feature classes (first-order statistics, texture matrices, shape descriptors) and merging extracted radiomic features with additional clinical or demographic predictors when available.

In the case of the MAMA-MIA dataset, the agent successfully merged the extracted radiomic features with external clinical and imaging variables provided by the dataset authors [25].

Across all datasets and scenarios, the agent successfully processed the images and segmentation masks, applied the requested image filters and feature selection, saved structured CSV files containing extracted features, with clear association to each patient ID, and saved parameter configuration logs for reproducibility. Visual inspection and manual validation confirmed that all requested radiomic features were extracted correctly, merged appropriately with external predictors when specified, and saved in the intended formats and output directories. No task failures were observed during the radiomic extraction experiments, and all agent outputs were consistent with those produced through direct tool execution. The output files generated during these experiments are provided in Supplementary Material Part D.

Image classifier agent

The Image Classifier Agent was evaluated using a series of datasets from the MedMNIST collection [21,22], each corresponding to different imaging modalities and classification challenges. For each dataset, the agent autonomously completed the full pipeline, which included model training, validation and testing based on the structured experimental prompts. Specifically, the agent successfully trained deep neural network architectures (ResNet18, ResNet34, ResNet50, ResNet101, ResNet152, VGG16, and InceptionV3), saved the best-performing model checkpoints as well as the final epoch model, and deployed the trained models for inference on independent test datasets. After inference, the agent evaluated the classification performance using standard metrics, including test set accuracy, macro-averaged precision, macro-averaged recall, and macro-averaged F1-score.

All training and inference tasks were completed without manual intervention or system failures across any dataset. The final evaluation results for each trained model on the respective MedMNIST test sets are summarized in Table 4.

Evaluation of multi-agent sequential workflows

To assess the system's ability to handle complex, multi-stage workflows autonomously, a series of multi-task queries were designed and executed. These tasks required the sequential activation and coordination of multiple specialized agents, with intermediate outputs serving as inputs for subsequent tasks.

In the first scenario (multi_task_prompt_1), the TotalSegmentator Agent was deployed to segment the spleen from a set of CT scans. The generated segmentation masks were then used by the Radiomic Feature Extraction Agent to extract spleen-specific radiomic features. Subsequently, the EDA Agent performed descriptive analysis of the extracted features, producing statistical summaries (Fig. 2). All masks, radiomic features, and EDA outputs were correctly saved to the designated directories, and no failures were observed at any stage.

In the second scenario (multi_task_prompt_2), the nnU-Net Developer and Implementer Agent was activated to segment multiparametric MRI (mpMRI) brain scans from the BraTS 2021 dataset. The Radiomic Feature Extraction Agent subsequently extracted radiomic features from the T1-weighted images across multiple filters (Original, Exponential, Wavelet). Exploratory Data Analysis was performed for each tumor subregion, and results were saved in structured subdirectories. All steps were successfully completed without user intervention.

The third scenario (multi_task_prompt_3) replicated a clinically

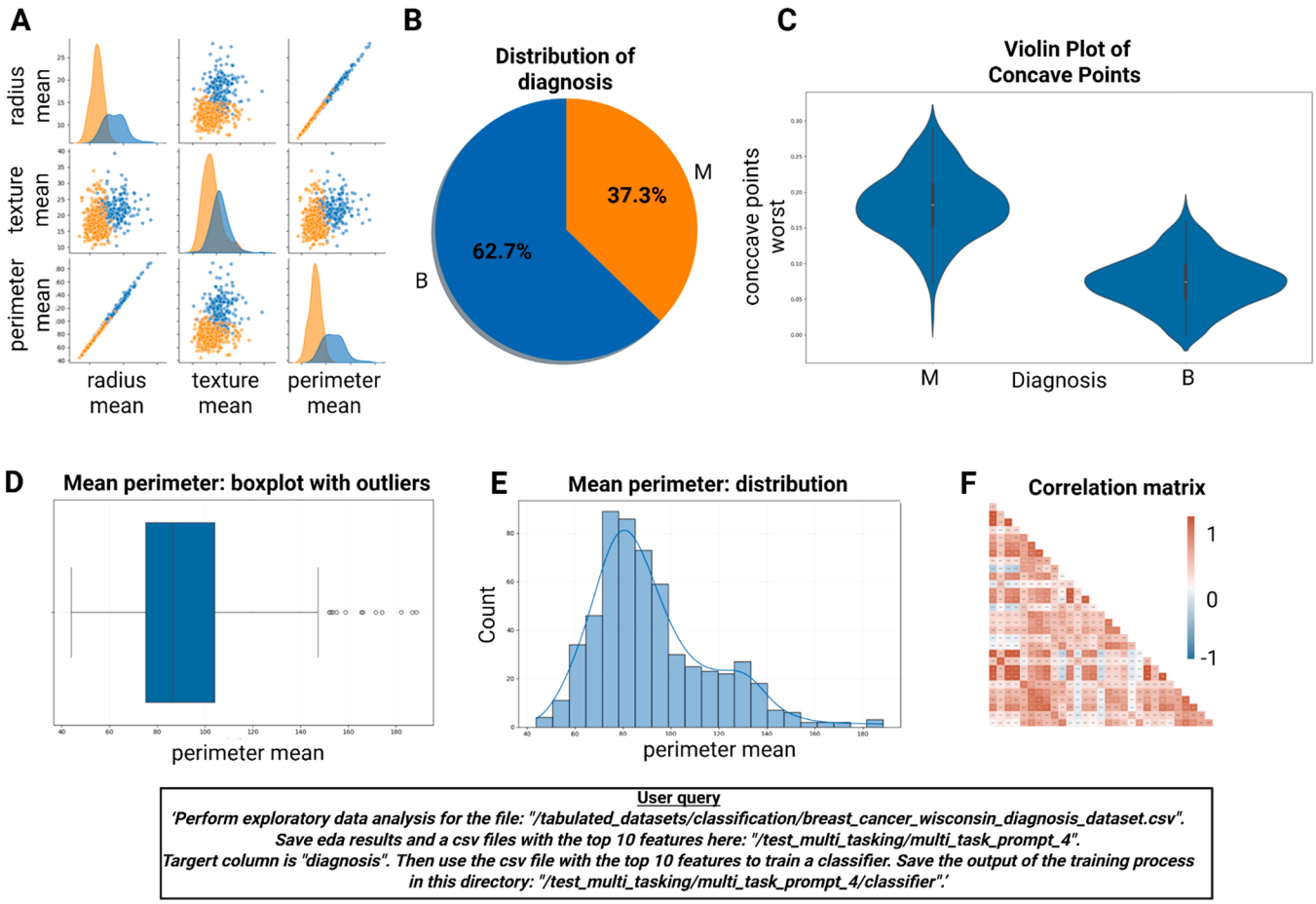

Fig. 4. Output generated from multi-task prompt 4, which combined exploratory data analysis, feature selection, and model training on a breast cancer classification dataset. This figure demonstrates some indicative results for selected dataset features. The EDA Agent produced a comprehensive set of visualizations, including pairwise correlation plots (A), pie charts demonstrating group distribution (B), violin plots for each feature per group (C – one feature is selected for demonstration purposes), boxplots and histograms for distribution and outlier analysis (D, E) and correlation matrices (F).

realistic research workflow (Fig. 3) using the newly released MAMA-MIA breast DCE-MRI dataset [25]. In this task, the Feature Importance Analysis and Selection Agent first identified the most predictive features for the binary classification target "pcr" (pathological complete response), a clinically important endpoint indicating complete eradication of detectable tumor following neoadjuvant chemotherapy. Subsequently, the Classifier Agent was deployed to train a model using the top 20 features. The resulting blended model achieved a mean (± standard deviation) cross-validation accuracy of 67.2 % (±4.5 %), a macro-averaged recall of 77.3 % (±7.6 %), a macro-averaged precision of 64.6 % (±4.7 %), and a macro-averaged F1-score of 70.1 % (±4.2 %). These results demonstrate the system's capability to autonomously process complex clinical datasets and produce clinically meaningful models.

In the fourth scenario (multi_task_prompt_4), a structured tabulated dataset (Breast Cancer Wisconsin) [16] was used. The system performed EDA, computed descriptive statistics, visualized key variables and distributions, and identified the ten most important features for breast cancer diagnosis (Fig. 4). Subsequently, it trained and evaluated a classification model based on these features, producing performance metrics, learning curves, and explainability plots (Fig. 5). All outputs were successfully generated and saved in the designated directories.

The fifth scenario (multi_task_prompt_5) tested a full end-to-end image classification workflow. The Image Classifier Agent successfully trained an InceptionV3 model on the PneumoniaMNIST (128 ×128) dataset, achieving a test set accuracy of 88.1 %, a macro-averaged precision of 91.3 %, a macro-averaged recall of 84.4 %, and a macro-averaged F1-score of 86.4 %. The AUC was 0.976. After training, the agent transitioned to perform inference on an independent test set, saving both the trained model and inference results without any manual intervention.

Across all multi-task scenarios, the system demonstrated operational autonomy, correctly managing agent switching, data transfer between tasks, error handling, and organized saving of intermediate and final outputs.

Discussion

An autonomous multi-agent system was developed to automate a wide range of medical AI tasks, including EDA, feature importance analysis, radiomic feature extraction and the development of segmentation and classification models across various imaging modalities. The system was developed to perform these tasks independently and generate comprehensive outputs and evaluation results for each completed process. By releasing mAIstro as an open-source framework, we aim to promote widespread adoption and support the implementation of standardized, reproducible, and well-practiced AI workflows in biomedical research and clinical applications.

The system was evaluated using diverse public datasets, including five tabular and fifteen medical image classification datasets, spanning applications such as breast cancer diagnosis, heart disease prediction, brain tumor segmentation, and pneumonia detection. Evaluation was conducted via natural language prompts simulating realistic, task-specific scenarios. At no point did the user instruct mAIstro which

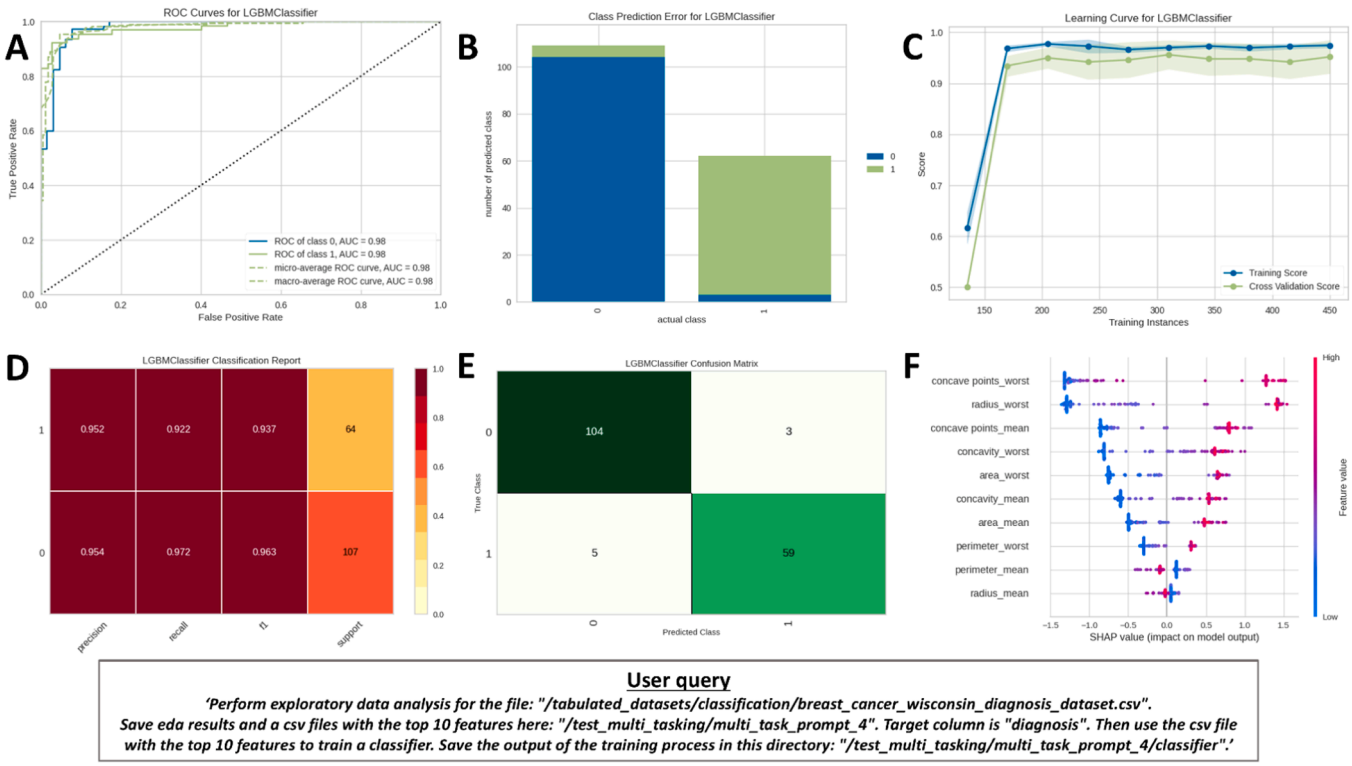

Fig. 5. Model evaluation results for multi-task prompt 4. Comprehensive model evaluation output is produced, including ROC curves (A), prediction error (B), learning curves (C), classification metrics (D), confusion matrix (E) and SHAP-based feature interpretation (F). All outputs were autonomously generated and saved by the agentic system.

agents or tools to use, nor how to solve each task. The user simply provided natural language prompts describing the desired outcome. mAIstro autonomously interpreted the request, devised a strategy, selected the appropriate agents and tools, determined their execution order, and coordinated the entire process end-to-end. A key strength of the system lies not only in its ability to perform individual tasks, such as EDA, feature importance analysis, or radiomic feature extraction, but also in its capacity to integrate these components into multi-step pipelines for the development, evaluation, and implementation of predictive AI models.

To evaluate the efficiency of mAIstro in autonomously developing medical image classification models, we compared our results with those reported in the MedMNIST publication [22]. Specifically, we performed one-to-one comparisons using the architectures and image sizes common to both works (ResNet-18 and ResNet-50 at 28 ×28 resolution). Out of 14 direct comparison instances, mAIstro outperformed the MedMNIST benchmarks in 11 cases. The largest improvement was observed in BreastMNIST (ResNet-50, 28 ×28), where mAIstro achieved 86.5 % accuracy compared to 81.2 % in MedMNIST publication [22] (+5.3 percentage points). The largest drop was in PathMNIST (ResNet-18, 28 ×28), where mAIstro reached 89.1 % compared to 90.7 % in MedMNIST (−1.6 percentage points).

To further assess the segmentation capabilities of mAIstro, we compared our results on the BraTS 2021 dataset with those reported in the literature [27]. Recent methods, including TCTNet [27], SegResNet [28], CoTR [29], TransBTS [30], CKD-TransBTS [31], UNETR [32] and SwinUNETR [33] report DSC values in the ranges of 90–93 % for whole tumor, 88–91 % for tumor core, and 79–89 % for enhancing tumor. In comparison, mAIstro achieved DSC values of 96 % for whole tumor, 95 % for tumor core, and 89 % for enhancing tumor. These results place our framework at or above the upper range of published state-of-the-art methods, demonstrating that mAIstro can autonomously deliver segmentation performance on par with the best existing approaches.

The framework is LLM-agnostic and was tested with a wide range of proprietary and open-source models, including GPT-4, Claude, DeepSeek, LLaMA, and Qwen. This flexibility allows offline operation in secure environments. Notably, mAIstro enables clinicians and researchers without coding expertise to build and evaluate AI models.

Although prior work has explored the use of LLM agents in healthcare, these studies are typically limited to narrow use cases such as diagnostic decision support or structured clinical simulations. Mehandru et al. [34] proposed a conceptual framework for the use of intelligent agents in clinical environments, emphasizing multi-step reasoning and decision-making, but without implementing or evaluating a functional system. Chen et al. [35] developed a multi-agent conversation (MAC) framework designed for rare disease diagnosis, demonstrating performance improvements through collaborative reasoning between agents, but again limited in scope to diagnostic tasks and structured medical interviews. Another framework, Tang et al.'s MEDAGENTS [36], focused on collaborative question answering across specialties but operated solely through language-based interactions, without the ability to process or analyze real-world data using external tools. To the best of our knowledge, this is the first multi-agent system designed for end-to-end data handling, EDA, radiomics extraction, model development, evaluation, and deployment across diverse data types in medicine, all orchestrated via natural language and tool-based interaction.

This study has limitations. The system's performance depends on the reasoning ability of the LLM, which can vary. While tool execution is deterministic, reasoning remains probabilistic and sensitive to prompt phrasing. To minimize the risk of hallucination, the framework is designed to operate under tool-grounded reasoning, where outputs are derived from validated tool executions. Clinical deployment would also require further validation under regulatory, ethical, and privacy constraints.

Conclusion

This study introduces mAIstro, the first open-source multi-agent

framework for autonomous, end-to-end AI development in medical imaging and tabular health data. Capable of performing EDA, radiomic extraction, segmentation, classification, and regression tasks, mAIstro enables non-programmers to interact with and evaluate AI pipelines using natural language. Moreover, experienced users can utilize the framework as a modular foundation, employing the full pipeline or specific agents independently, developing new tools, and extending the system to meet specialized research or clinical needs.

CRediT authorship contribution statement

Michail E. Klontzas: Writing – review & editing, Visualization, Validation, Supervision, Project administration. **Eleftherios Tzanis:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethics statement

Ethics committee approval was not required, as all data used in this study are publicly available, designated for open research use, and appropriately cited in the manuscript.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: M.E.K. serves as advisory editor and E.T. serves as junior editor for this journal but were not involved at any step of the review and decision process. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ejrai.2025.100044.

Data availability

The code developed for this study is openly available at: https://github.com/eltzanis/mAIstro

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